REVIEW

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Engineering

Research and development of helium-xenon Brayton cycle technology: A review

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Abstract: Helium-xenon Brayton cycle systems have significant potential as the energy conversion system for small-scale reactors in remote land, deep-sea, and space applications due to a range of advantages, including high cycle efficiency, compact system structures, and chemical stability. The objective of this review is to provide a comprehensive understanding of the helium-xenon Brayton cycle system based on the projects and researches. First, the basic information and development history are introduced, and a series of typical designs are summarized. Then, the system configurations, cycle parameter analysis and optimization are discussed. Next, the key components are classified, such as turbine, compressor, and heat exchanger. Moreover, the dynamic processes and control strategies are introduced in different conditions. Finally, the deficiency and prospect of current research are presented. The review covers the representative helium-xenon Brayton cycle systems, which could provide a reference for promoting the development of energy conversion systems.

Keywords: helium-xenon Brayton cycle, cycle analysis, heat exchanger, dynamic research

INTRODUCTION

In the current context of escalating energy demands and increasing global energy security concerns, the pursuit of safe, efficient, and environmentally friendly energy sources is crucial for addressing the challenges posed by climate change and achieving sustainable development. Fourth-generation nuclear power generation technologies offer enhanced levels of safety and reliability, with energy conversion systems in nuclear reactors playing a pivotal role in shaping the future of the power generation sector. Brayton cycle systems employing helium or helium-xenon mixed gas as the working fluid possess a range of advantages, including high cycle efficiency, compact system structures, exceptional compression performance, and chemical stability. These systems are particularly well suited for dynamic thermoelectric conversion in small-scale reactors, offering promising prospects in fields such as space nuclear power, deep-sea nuclear power, and mobile nuclear power in remote regions.

Since the mid-20th century, nations such as the United States, the Soviet Union, France, Germany, and Japan have successively engaged in designing and researching helium or helium-xenon Brayton cycles [1]. The helium or helium-xenon Brayton cycle can serve as an energy conversion system for land-based high-

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temperature gas-cooled reactors. International research on high-temperature gas-cooled reactors commenced in the 1960s, resulting in the development of experimental power stations in the United Kingdom, the United States, and Germany. In the 1990s, countries including Japan, China, and South Africa conducted relevant studies on modular high-temperature gas-cooled reactors incorporating Brayton cycle systems. Concerning space Brayton cycles, the United States and the Soviet Union have embarked on separate space nuclear energy projects since the 1960s, focusing on the design and experimental work of space reactors utilizing helium-xenon Brayton cycle systems. Noteworthy programs in the United States include the SP-100 plan and the Brayton Rotating Unit (BRU) plan [2,3]. Related research has also been undertaken in France, involving the design of various Brayton cycle systems coupling of different reactors, such as the Nuclear Electro-Tug for Interorbital Transfer (ERATO) plan. Following the dissolution of the Soviet Union, Russia gradually shifted its primary research direction toward the design of high-power dynamic energy conversion systems. In China, esteemed educational institutions and research organizations such as Tsinghua University, Xi'an Jiaotong University, and the China Aerospace Science and Technology Corporation (CASC) have conducted extensive work on land-based helium Brayton cycles and space helium-xenon Brayton cycles [4].

In the 1960s, National Aeronautics and Space Administration (NASA) initiated the BRU project. In 1968, NASA developed a ground test prototype for the Brayton cycle using a working fluid composed of helium-xenon gas with a molecular mass of 83.8 g mol^{-1}. The prototype had a power range of 2.25–10.5 kW_e and underwent testing for over 38,000 h. Building upon the BRU program in 1974, NASA launched the mini-BRU program to design a more compact and efficient thermoelectric conversion rotating unit. By 1978, a ground test prototype with higher cycle efficiency, ranging from 500 to 2100 W_e, was developed and applied in the Brayton Isotope Power System (BIPS), which has been tested for over 1000 h [4].

In 2002, NASA announced a new space nuclear power program to promote and advance the development of space nuclear power sources, including propulsion and nuclear reactor power system development for the Jupiter Icy Moons Orbiter (JIMO). In 2003, the project was renamed the "Prometheus" program, designing a 200 kW_e, 1 MW thermal power reactor unit [5]. Various reactor core types were evaluated and matched with different thermoelectric conversion methods, including closed Brayton cycles, Stirling cycles, and thermophotovoltaic systems. The final design adopted a high-temperature gas-cooled fast reactor coupled with a direct Brayton cycle. The core outlet average temperature was 1150 K, and the reactor and shielding mass ranged from 3000 to 5000 kg, with a spacecraft total mass ranging from 7500 to 11,000 kg. The Prometheus program was terminated in 2006, but this resulted in the design of a representative gas-cooled single-loop dual-machine cycle system.

In 2007, King and El-Genk [6,7] designed the Submersion Subcritical Safe Space Reactor (S4) to enhance the safety of space missions and ensure the safety of reactors and space systems in the event of launch abort accidents or operational failures. The S4 reactor was based on a segmented thermoelectric pile (HP-STMC) and an Alkali Metal Thermal-to-Electric Converter (AMTEC) integrated space reactor power system. Three helium-xenon Brayton cycles were used as the energy conversion system, with a total power generation capacity of 100 kW_e. The cylindrical core was divided into three hydrodynamically independent but neutron-physically and thermally coupled core regions, each connected to a separate helium-xenon closed Brayton cycle system and its corresponding heat dissipation device. This design demonstrated the feasibility of multiple power cycle coupled reactor systems, avoiding single-point failures and providing redundancy for the thermoelectric conversion process. In the 1960s, the Soviet Union initiated a space nuclear power

program with a focus on developing space nuclear reactor power sources and nuclear thermal propulsion technology. The primary methods of space reactor thermoelectric conversion include static conversion using a temperature difference and thermal ions, with a series of typical reactor designs such as BUK and TOPAZ [8].

Following the dissolution of the Soviet Union, Russia transitioned toward the development of higher-power space reactor dynamic conversion power systems and high-power space nuclear-powered spacecraft. In 2009, Russia announced a development plan for a megawatt-class space nuclear-powered spacecraft for space missions. This spacecraft employed nuclear thermal propulsion utilizing a helium-xenon gas-cooled fast reactor coupled with a closed Brayton cycle for thermoelectric conversion. Waste heat dissipation was achieved using a droplet radiative cooler. The spacecraft provided an electric power output of approximately 0.8–1 MW_e with a system efficiency of 28.6%.

This study focuses on summarizing and synthesizing aspects related to the working fluid characteristics, parameter analysis and optimization, component design, dynamic characteristics, and control strategies of helium and helium-xenon Brayton cycle systems.

SYSTEM ANALYSIS

Characteristics of helium and xenon

In terms of thermodynamic performance, helium is considered one of the optimal working fluids for the Brayton cycle, especially in land-based nuclear reactor Brayton cycle systems. Helium, as an inert gas, exhibits stable properties, favorable specific heat capacity, and thermal conductivity. These properties contribute to a reduction in heat exchanger volume, resulting in a compact structure and making it suitable as a coolant for small-scale nuclear reactors. However, the low molar mass of helium makes it prone to leakage from the cycle, and its compressibility is relatively poor, leading to a greater number of stages needed for helium turbines and compressors, ultimately increasing the total volume of the system. Additionally, boundary layer losses contribute significantly to increased aerodynamic losses in the system.

On the other hand, xenon, as an inert gas, has a larger molecular mass, which enhances its compressibility. However, its heat transfer performance is comparatively inferior, resulting in larger heat exchanger volumes. By blending helium and xenon in specific proportions, it is possible to strike a balance between their respective advantages, achieving both excellent heat transfer performance and compressibility. This blend is well suited as a working fluid for Brayton cycle systems in compact space-based microreactors, where volume constraints are crucial [4].

Several numerical and experimental researches on heat transfer characteristics of helium-xenon mixture are conducted (Table 1). Taylor *et al*. [9] conducted a series of experiments on helium-xenon gas mixture with Prandtl numbers in the range of 0.18–0.7, which is the earliest convective heat transfer experiment of helium-xenon in the open literature. Qin *et al*. [10] and Wu *et al*. [11] conducted a series of numerical and experimental investigation on the helium-xenon gas mixture. The flow and heat transfer characteristics are investigated during the Reynolds number range of 4000–30,000. Yang and Shi [12,13] developed a code to predict the physical properties of helium-xenon mixture based on Chapman-Enskog theory and analyzed the effects of physical properties change on Brayton system performance. But as for the Brayton cycle system,

Table 1 Physical properties of air, helium, xenon, and He-Xe mixture

Parameter	Air	Helium	Xenon	He-Xe
Molecular weight (g mol^{-1})	28.96	4	131.29	40
Specific heat capacity at constant pressure C_p (J kg^{-1} K^{-1})	1006.6	5193.2	160.1	519.8
Specific heat ratio	1.4018	1.6665	1.6773	1.6680
Kinetic viscosity (μPa s)	18.490	19.846	22.985	24.960
Thermal conductivity (W m^{-1} K^{-1})	0.0259	0.1553	0.0055	0.0279
Density (kg m^{-3})	1.184	0.164	5.395	1.636
Sonic velocity (m s^{-1})	346.3	1016.4	177.0	321.4

the experimental researches on the thermohydraulic characteristics of helium and xenon are still in need to explore.

In this part the characteristics of helium and xenon are introduced. The current research showed that the heat transfer and flow characteristics of helium-xenon mixture should be considered comprehensively to match the requirements of diverse scenarios. By now the experimental investigations are still limited by the high cost of inert gases. Substitute gas with similar properties would be an available way in the field of heat exchanger or turbomachinery experiments.

Cycle layout analysis

Based on the different application scenarios, the appropriate configurations for helium and helium-xenon Brayton cycles can vary. The use of a reheating cycle with intercooling is more suitable for the construction of large land-based nuclear power stations considering the system size and safety implications of the components. Conversely, a simple recuperating cycle is appropriate for applications in land-based mobile reactors and space reactors due to the restrictions on the load of vehicles and spaceships [4,14]. The addition of intercooler serves to decrease the temperature of the working fluid during compression, which makes it easier to compress and reduces compressor power consumption. However, the use of heat exchangers also results in increased system pressure losses without significantly improving the cycle efficiency. As for the reheater, it can also improve cycle efficiency, but will increase volume and mass. Additionally, the addition of components may adversely impact system reliability. Therefore, if the system has strict requirements in terms of mass and volume, intercoolers and reheaters are not essential.

Holos-Gen, a company based in the United States, has designed a series of Holos microreactor heat-to-electricity conversion systems with varying power levels [15–17]. A system with a 1 MW electrical output is primarily intended for space and highly mobile applications. This system utilizes a high-temperature gas-cooled reactor coupled with four direct helium or supercritical carbon dioxide (S-CO_2) Brayton cycle systems, each equipped with intercooling. Moreover, the proposed approach can be combined with an organic Rankine cycle (ORC) bottoming cycle, optimizing the overall cycling efficiency from 45% to 60%. With its ability to be mounted on specialized helicopters, this mobile reactor system provides a rapidly deployable power generation solution for disaster-stricken areas experiencing power outages (Figure 1).

The UK-based company Urenco, in collaboration with the University of Manchester and Delft University of Technology, has jointly designed the U-Battery, a mobile microreactor [17]. Originally, a high-temperature gas-cooled reactor coupled with a direct helium Brayton cycle was utilized. However, due to design issues

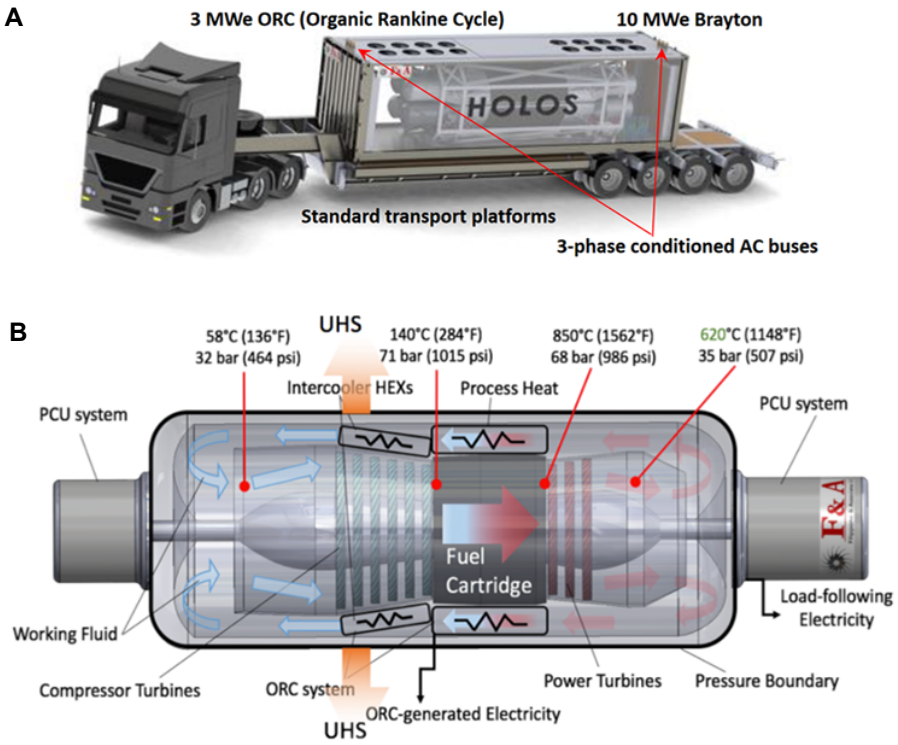


Figure 1 **(A)** Holos mobile generator scheme. **(B)** Holos generator cycle system scheme [15,16]. Copyright 2018, HolosGen LLC.

with helium turbines, the energy conversion system was later modified to include an indirect nitrogen Brayton cycle.

Toshiba Corporation in Japan developed the MoveluX microreactor for application in distributed power grids (Figure 2). By combining a sodium heat pipe reactor with a recuperating helium Brayton cycle system, this reactor provides a thermal power output of 10 MW. Construction of the MoveluX reactor is expected to begin around the year 2030 [18].

Iregui and Isabel [19] designed a hybrid microreactor system with a solar-assisted helium Brayton cycle for Nazareth, La Guajira, Colombia. This system is intended to provide continuous power to rural areas. It consists of a microreactor and a solar collector utilizing a helium recuperating Brayton cycle as the energy conversion system. Based on the simulation results, this system has the capacity to deliver 55 GWh of electricity annually to rural areas, offering advantages such as a small footprint, sustainable power generation, and low-carbon cleanliness.

The X-Energy Corporation developed the Xe-Mobile microreactor energy conversion system, which is currently in the conceptual design phase. The plan entails combining a high-temperature gas-cooled reactor with a helium Brayton cycle (Table 2).

For space missions requiring high power and long durations, the utilization of high-temperature gas-cooled reactors or heat pipe reactors combined with a helium-xenon Brayton cycle is preferred. Extensive design and experimental investigations on this topic have been conducted by the United States and the Soviet Union since the 1970s. Notable plans for space Brayton cycle systems include the United States Prometheus

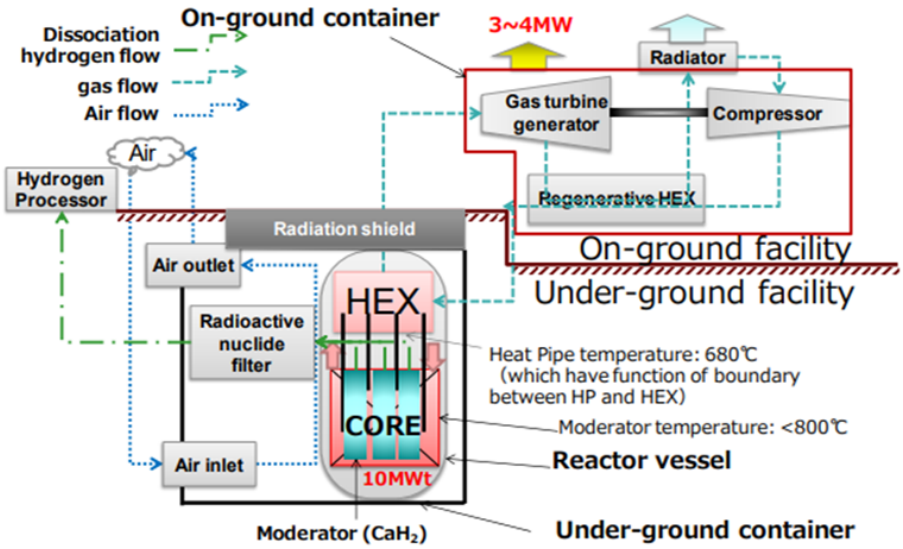


Figure 2 MoveluX system scheme [18]. Copyright 1995–2024, TOSHIBA CORPORATION.

Table 2 Land-based Brayton cycle projects

Project	Nation	Reactor	Medium	Thermal (MW)	Electric (MW)	Efficiency (%)
Holos	US	High temperature reactor (HTR)	He	22	–	44.63
Xe-mobile	US	HTR	He	–	1–5	–
U-battery	UK	HTR	He	10 × 2	4 × 2	–
UNSCMMR1	Columbia	HTR	He	15	–	–
MoveluX	Japan	Heat pipe reactor (HPR)	He	10	3–4	–
StarCore	Canada	HTR	He	–	10 × 2	–
SIMONS	China	HTR	He-Xe	20	–	40
Mi-HTR	China	HTR	He	5	–	34.4
MINION	US	HTR	He	1	–	–

program, the Russian transport and energy module (TEM) program, and the French ERATO program, among others. The United Kingdom has also considered helium Brayton cycles in the design of the Skylon spaceplane, while China has conducted conceptual designs for megawatt-class space reactors (Table 3).

To ensure the normal and safe operation of space missions, redundancy loops are commonly integrated into space energy conversion systems. These loops are arranged symmetrically or distributed evenly in thirds, forming dual-loop or multiloop cycles.

In 2002, the NASA in the United States announced a new space nuclear initiative to promote and advance the development of space nuclear power sources. In 2003, it was renamed the “Prometheus” program, which focused on the propulsion and nuclear power system development for the JIMO. The program evaluated various reactor types, thermoelectric conversion methods, and system configurations. Among them, four multiloop cycle configurations were designed, including single-loop, dual-loop, triple-loop, and quadruple-loop Brayton cycles [5,20]. Each individual loop constitutes a simple recuperating cycle, with the operating power being adjustable to provide redundancy for safe operation (Figure 3).

Table 3 Space Brayton cycle projects

Project	Nation	Reactor	Medium	Electric (kW)	Efficiency (%)
SP-100	US	Lithium fast reactor (LFR)	He-Xe	10–100	20.7
Prometheus	US	HTR	He-Xe	200	20
SAFE	US	HPR	He-Xe	100	25
S4	US	HTR	He-Xe	93.6	21.4
Skylon	UK	–	He	–	–
TERRA	Brazil	HPR	He	–	–
TEM	Russia	Sodium fast reactor (SFR)	He-Xe	1000	–
		SFR		20	16.8
ERATO	France	HTR	He-Xe	20	17.3
		LFR		20	17.6
Megawatt space Nuclear reactor	China	HPR	He-Xe	1000	–

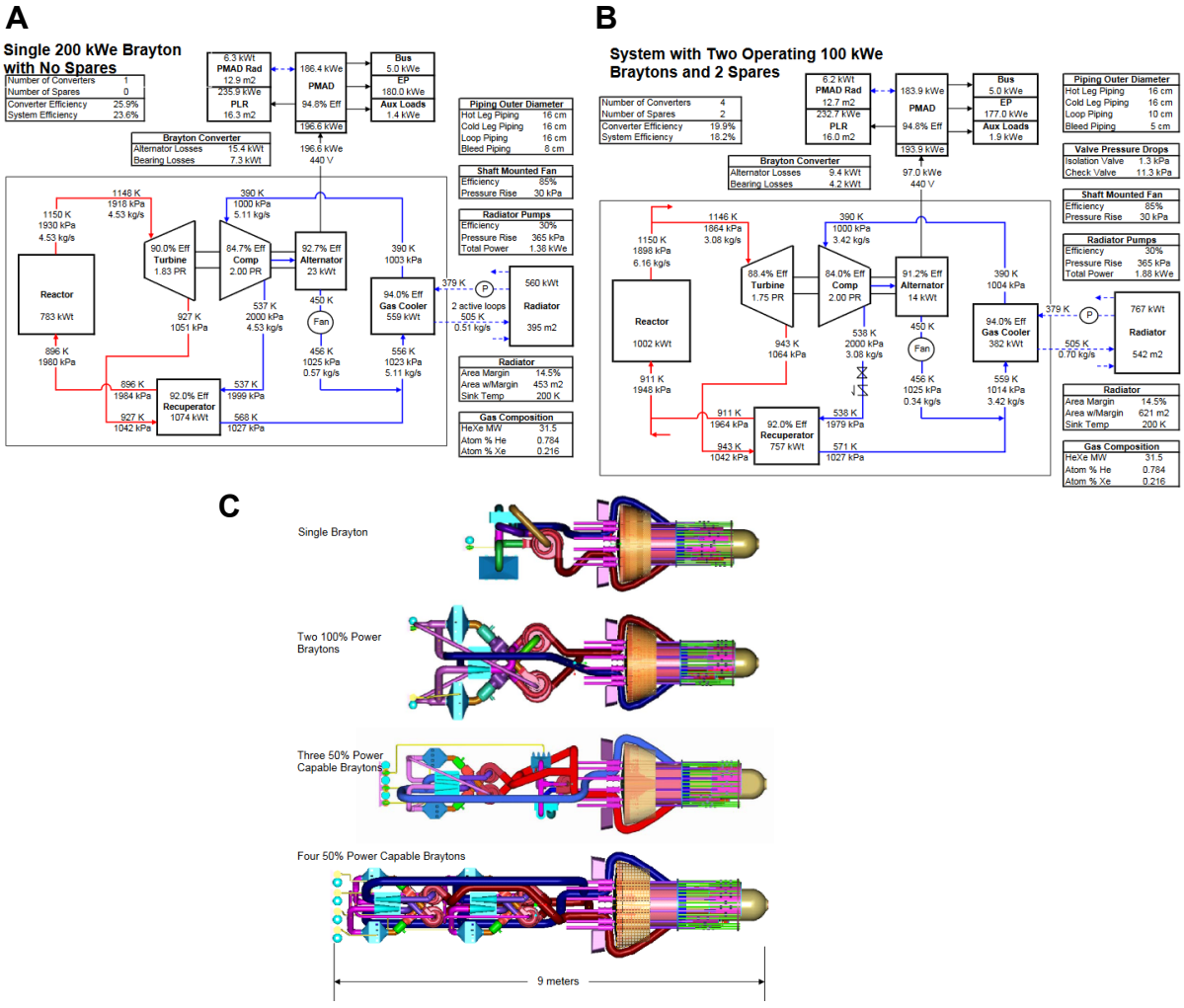


Figure 3 (A) Prometheus single-loop cycle. (B) Prometheus dual-loop cycle. (C) Prometheus system arrangement options [5]. Copyright 2007, AIP Publishing.

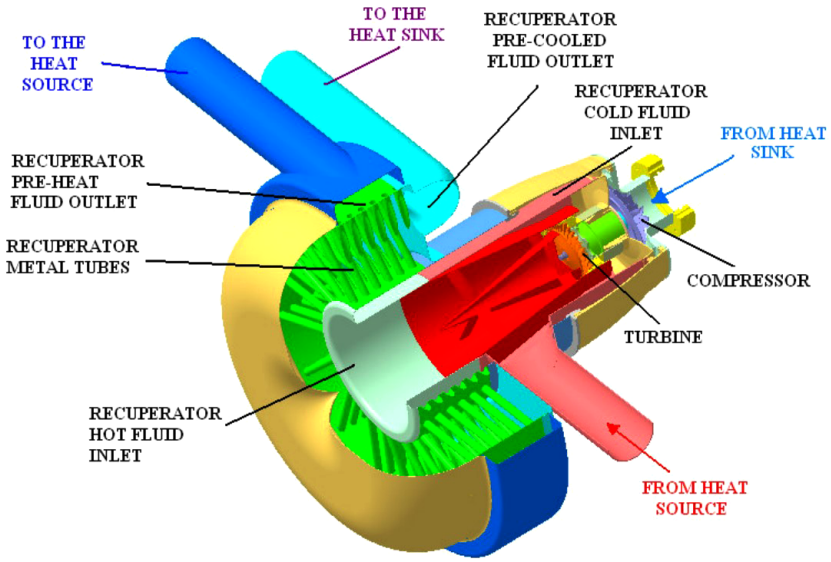


Figure 4 TERRA system scheme [21].

Based on the Prometheus project, Guimarães *et al*. [21] developed and constructed the TEcnologia de Reatores Rápidos Avançados (TERRA) project. This project involves an energy conversion system that combines a heat pipe microreactor with a recuperating helium Brayton cycle (Figure 4).

In 2007, King and El-Genk [6,7] proposed the design of a novel S4. The S4 reactor has a system power of 100 kW_e and is based on the design of an HP-STMC and an AMTEC integrated space reactor power system. The reactor core is cylindrical and divided into three hydraulically independent yet neutronically and thermally coupled reactors. Each reactor is connected to a separate closed Brayton cycle system, utilizing helium-xenon gas cooling and corresponding heat dissipation devices. This design validates the feasibility of coupling multiple power cycles to a reactor, providing redundancy for potential single-point failures in the reactor core cooling and thermoelectric conversion processes (Figure 5).

Juhasz [22] proposed a power station development concept for lunar surface colonization (Figure 6). The concept involves the use of a high-temperature gas-cooled reactor coupled with two parallel-arranged closed simple helium Brayton cycles. Heat dissipation is achieved through a vertical heat pipe radiator. Juhasz noted that, in terms of heat dissipation, two symmetrically arranged 5 MW_e cycles offer advantages over a single 10 MW_e cycle, for this system design would simplify the radiator configuration by separating one big radiator into two smaller ones. Moreover, the two-system arrangement would improve system safety in operation [22,23].

Zhang *et al*. [24,25] developed a design proposal for a megawatt-class space reactor system (Figure 7) and conducted simulation calculations. The reactor has a power output of 3.2 MW with a cycle efficiency of 31.8%. Heat is subsequently exported through lithium heat pipes to a dual-loop helium-xenon closed Brayton cycle. The cold end is cooled by a sodium-potassium loop and then transferred to a potassium heat pipe radiator for radiative cooling.

The following table lists several multiloop Brayton cycle systems (Table 4).

In addition, a portion of the working fluid is extracted from the compressor outlet for air bleeding, which is

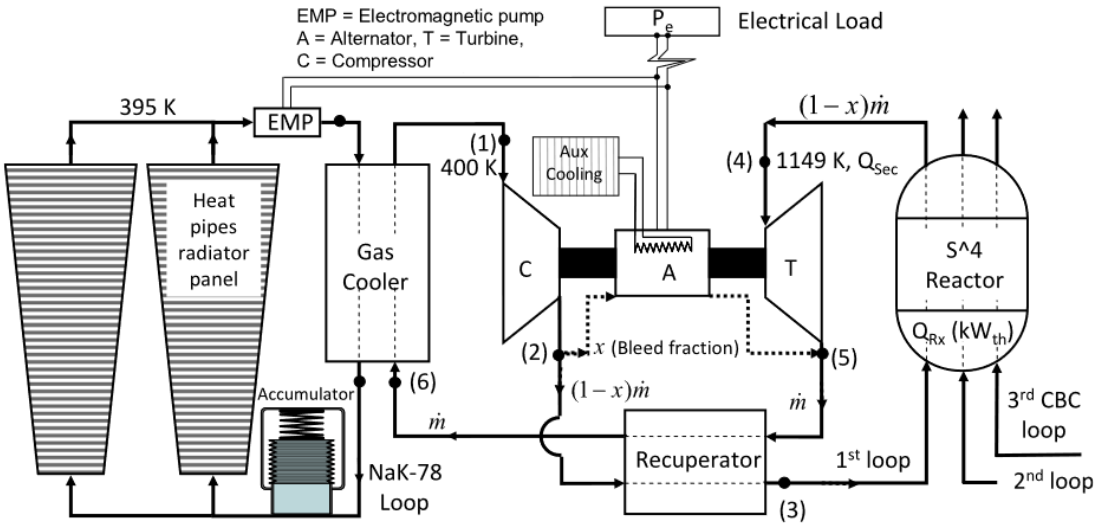


Figure 5 S4 system scheme [7]. Copyright 2006, Elsevier B.V.

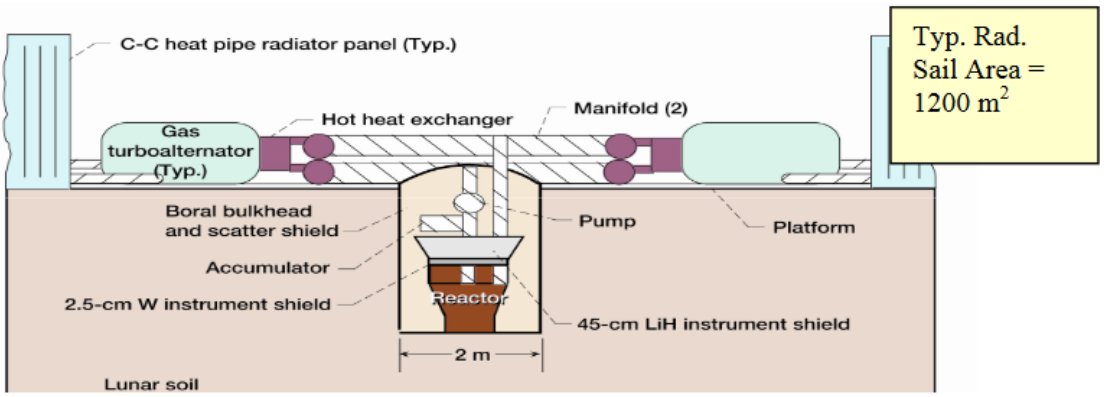


Figure 6 Helium cycle power station for lunar colonies [22].

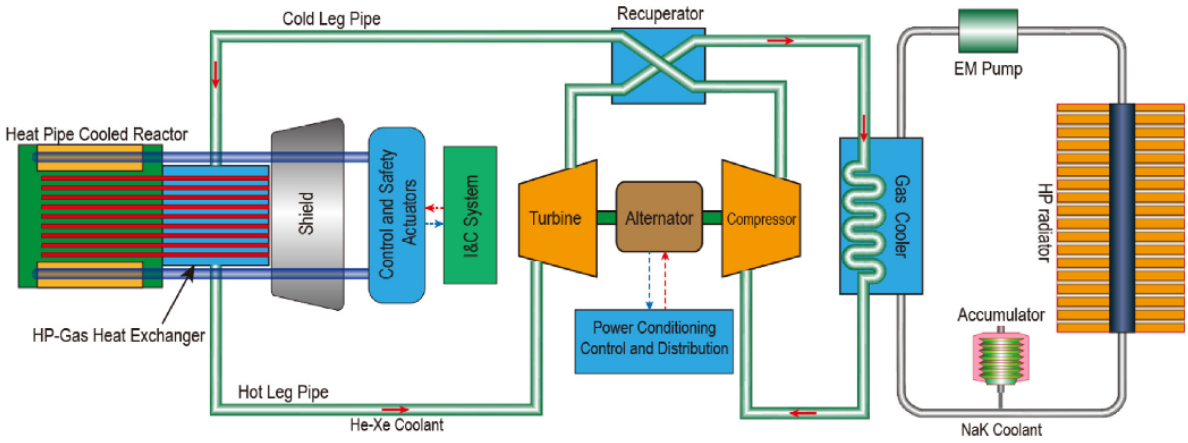


Figure 7 Megawatt-level space reactor system [24].

Table 4 Multiloop Brayton cycle projects

Project	Nation	Reactor	Medium	Loop
ERATO	France	SFR, HTR, LFR	He-Xe	2
CCGT for lunar colonies	US	HTR	He	2
Prometheus	US	HTR	He-Xe	2/3/4
S4	US	HTR	He-Xe	3
SNRPS	China	HTR	He-Xe	2
OMEGA	China	HTR	He-Xe	4

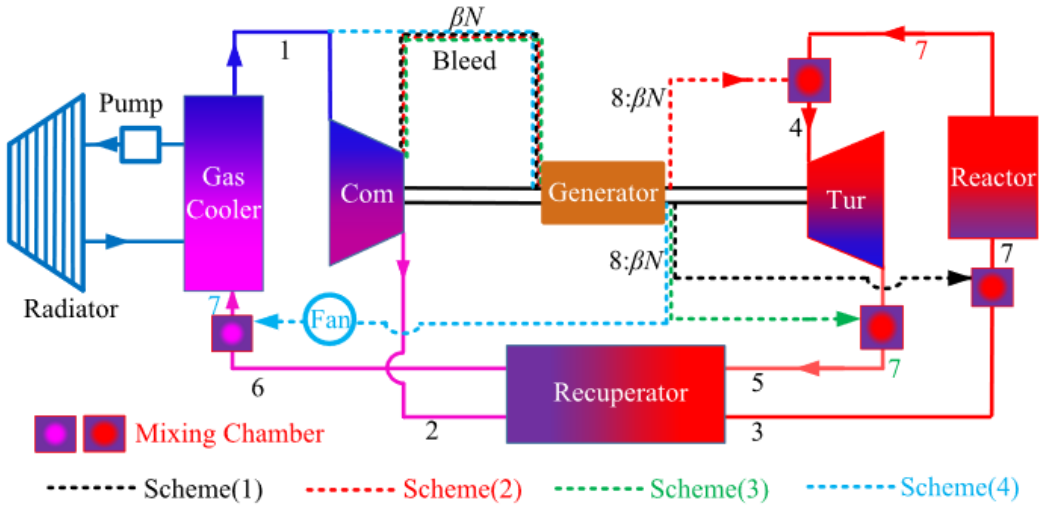


Figure 8 Bleed scheme of cooling fluid [27]. Copyright 2021, Elsevier Ltd.

used for cooling the bearings and generator. Configurations involving air bleeding can utilize waste heat from bearings and generators, eliminate the need for a separate cooling system, enhance system reliability, further reduce system volume and mass, and increase overall efficiency.

King and El-Genk [7] and El-Genk [26] introduced 2% air bleeding in the closed Brayton cycle corresponding to the S4 reactor, which is employed for cooling the shaft and generator [7,26]. This air is drawn from the compressor outlet and mixed with the main gas at the turbine inlet. Ma *et al*. [27] conducted relevant research on cooling air bleeding for helium-xenon cycles in space reactors. They designed four different cooling air bleeding schemes (Figure 8), analyzed the impact of bleeding air on key components and cycle performance, and optimized and compared the four schemes under the limitation of the temperature pinch point in the heat exchanger. The results indicated that bleeding air to the reactor and turbine inlets could enhance cycle efficiency.

Huang [28] established and optimized four configurations for helium Brayton cycle systems: no recuperating cycle, simple recuperating cycle, simple intercooling cycle, and intercooling with recuperating cycle. The focus of the study was to determine the impact of intercooling and reheating on the cycling performance. The results demonstrated that reheaters greatly enhance cycle thermal efficiency and reduce the required pressure ratio for achieving optimal cycle efficiency. Although intercoolers can reduce compressor power consumption, their effectiveness is influenced by pressure loss. Therefore, the decision to incorporate

intercooling needs to consider factors such as overall cycle performance, system complexity, and construction costs.

In a study conducted by Bae *et al*. [29], a 20 MW_{th} SM-HTGR helium Brayton cycle system was researched considering three configurations: simple recuperating, simple intercooling, and double intercooling. The addition of intercooling improves the cycle performance but does not necessarily enhance the system value. While the cycle pressure increases and electrical arcing decreases with the inclusion of intercooling, this may also lead to aerodynamic losses, resulting in a decrease in turbine efficiency. A comparison between the helium Brayton cycle and S-CO_2 Brayton cycle was performed, and the results suggested that under similar conditions, S-CO_2 outperforms helium.

Wright *et al*. [30] studied the impact of interstage heating and cooling (IHC) on helium cycle systems. They extensively analyzed various IHC configurations (1T2C, 3T3C, 3T6C, etc., where T means turbine and C means compressor), single-axis and multi-axis arrangements, and compact and distributed designs. They also compared the performances of helium and 70%–30% helium-argon mixed gas. The results revealed that the 3T6C configuration of the IHC cycle had similar volume dimensions to those of a Rankine cycle system with the same power level. Its advantages lie in its insensitivity to pressure drops and high heat exchanger efficiency.

Postlethwait *et al*. [31], in the context of the Prometheus project, conducted a comparative study on space helium Brayton cycles using direct and indirect cycles. The indirect cycle, which utilizes heat exchangers to isolate the first and second loops, enhances safety. However, the resulting heat exchanger mass is approximately 32%–57% of that in the direct cycle, without significant improvements in efficiency.

In this part different cycle layout designs of helium-xenon Brayton cycle systems are introduced, including land and space scenarios. Configurations, cycle loops and arrangements are analyzed in the researches. Most of the researches are based on numerical simulation and model programming, and experimental researches are mostly based on non-nuclear test base in case of accident. In the future more system experiments including reactors can be explored.

Cycle performance analysis and optimization

Improving cycle efficiency is a perpetual topic for cycle systems. Additionally, for different application scenarios, optimization of structural parameters such as system size and mass is necessary. In land-based helium or helium-xenon Brayton cycle systems, cycle efficiency enhancement can be achieved by adopting more complex configurations, strengthening heat exchangers, including reheaters, and coupling with bottoming cycles. However, in space helium-xenon Brayton cycle systems, the focus of system optimization is primarily on goals such as the power density and power-to-mass ratio to ensure compactness and safety in reactor systems.

Xue [32], referencing the S4 reactor, established a small helium-xenon closed-loop cycle system with system efficiency and mass as optimization goals. The study showed that when the recuperator effectiveness exceeds 96%, the surface area of the heat exchanger and the overall system mass increase rapidly, which negatively impacts the control of the system mass.

El-Genk and Tournier [33] and Tournier *et al*. [34] conducted extensive studies on the thermophysical properties of helium, other inert gases, nitrogen, carbon dioxide, and mixtures of different ratios of gases. The

results indicated that when helium is mixed with xenon or krypton, the heat transfer performance initially increases and then decreases, while the pressure loss in heat exchange components gradually increases. A mixture of 40 g mol^{-1} helium-xenon is suitable for space cycle systems, a mixture of 15 g mol^{-1} helium-xenon is suitable for land-based high-temperature gas-cooled reactors, and helium-krypton with a molecular weight of 22 g mol^{-1} also exhibits good heat transfer performance. You *et al*. [35] compared the transport characteristics and heat transfer capabilities of helium, nitrogen, carbon dioxide, and their binary mixture gases by conducting studies on coupled closed Brayton cycles. The results showed that a mixture of 20 g mol^{-1} helium-CO_2 has a higher heat transfer coefficient than helium, reaching only 18%–20% of helium at 1200 K, which can significantly reduce the load on turbine machinery, reaching only 29%–34%.

Invernizzi and Marcoberardino [36] analyzed the thermal characteristics of closed Brayton cycles under the influence of actual gas properties as well as the compatibility and thermal stability of high-temperature materials. The inclusion of mixed gases of carbon dioxide and inert gases expands the application range of closed Brayton cycles, allowing for continuous changes in the critical temperature of the working fluid and, in some cases, improved the thermodynamic cycle efficiency.

Li *et al*. [37,38] modeled a space helium-xenon Brayton cycle system using Fortran language and analyzed the effects of changes in the composition of the mixed working fluid on gas physical properties and thermodynamic properties. Traditional land-based Brayton cycle systems utilize the operating modes of maximum efficiency; however, in the case of limited mass and volume of space reactors, Li proposed an optimization mode based on maximizing the cycle ratio of specific work. Two optimization modes were established in Fortran language and compared. The specific work optimization mode offers lower mass flow and pressure, which can reduce the flow area of the heat exchangers. But it has higher compressor pressure ratio, which will bring challenges in the turbomachinery design. As a result, the two modes should be considered together and the final optimized solution could be chosen based on engineering requirements.

Liu *et al*. [39] utilized the Non-dominated Sorting Genetic Algorithm II (NSGA-II) to minimize the total system mass by optimizing key parameters of system components. The Garson Algorithm was employed for sensitivity analysis of various components of the helium-xenon Brayton cycle. Calculations demonstrated that a 4% increase in the turbine inlet temperature from 1150 K could result in a 6% reduction in system mass.

Ribeiro *et al*. [40] conducted an analysis and optimization of the heat exchanger size and mass in a closed regenerative Brayton cycle system coupled with a space heat pipe reactor. By maintaining the overall volume constant, they optimized the system efficiency by altering the volume ratios of different heat exchangers or by keeping the overall volume constant while optimizing efficiency. The results showed that increasing the heat exchanger efficiency can enhance the cycle efficiency, but consideration must be given to the heat exchanger mass. Additionally, improving the variable efficiency of the compressor and turbine is considered a crucial opportunity to enhance system performance. Increasing the radiator panel area by 25% could result in an 8.33% efficiency improvement, offering significant potential for enhancing the overall system efficiency.

Biondi and Toro [41] modeled and analyzed a solar collector coupled with a helium-xenon closed recuperating Brayton cycle (Space Solar (thermal) Dynamic Systems Closed Brayton Cycle SDS-CBC). The optimization focused on parameters such as the compressor inlet temperature, pressure ratio, and receiver diameter with the objective of minimizing the system mass. The simulation results revealed a 20.83%

reduction in the system weight, a 7.41% improvement in the cycle efficiency, and the potential for achieving the target of a 30 kg kW^{-1} power-to-mass ratio. However, further research is needed. This study can also serve as a reference for closed regenerative Brayton cycle systems in microreactors.

In summary, when considering helium and helium-xenon Brayton cycle systems adapted for microreactors, optimizing cycle efficiency is not the sole goal. Comprehensive optimization should also take into account overall system compactness, including objectives such as volume, mass, and other factors. The safety and long-term operation of the system can be considered as well.

KEY COMPONENTS DESIGN AND ANALYSIS

Turbine and compressor

Research on the design of turbines and compressors for helium and helium-xenon cycles can be traced back to the 1960s with the development of direct Brayton cycle systems for high-temperature gas-cooled reactors. In 1968, Germany conducted research and development on closed Brayton cycles, constructing and operating an experimental combined heat and power plant with a thermal power of 160 MW, supplying 50 MW of electricity and 53 MW of district heating. Another project involved the High-Temperature Helium Test Power Plant (HHV), which tested the turbine and compressor components. In the late 1980s, relevant research faced setbacks owing to nuclear accidents and antinuclear movements. After the end of Cold War, researches on high-temperature gas-cooled reactors were put more focus on the modular design. In the 1990s, the United States and Russia jointly initiated research on Modular High-Temperature Gas-Cooled Reactor (MHTGR) coupled with helium Brayton cycles. In 1994, South Africa developed the Pebbled Bed Modular Reactor (PBMR). Though the development was suspended in 2010, it shows great impact in modular design [42,43]. In 2001, Japan started to develop the Gas Turbine High Temperature Reactor of 300 MW_e nominal capacity (GTHTR300). Sato *et al*. [44] and Yan *et al*. [45] improved the GTHTR300 design to achieve an efficiency of 50%. In 2003, Tsinghua University developed the 10 MW High Temperature Gas-cooled Test Reactor (HTR-10), which is the first high-temperature gas-cooled reactor test module in China [1]. Based on it, the High Temperature Reactor-Pebble bed Modules (HTR-PM) is successfully put into operation in Shandong Province, which is the first fourth generation nuclear power plant in the world.

In the initial stages, the design of helium compressors drew inspiration from air compressors. Escher Wyss developed four helium compressors between 1961 and 1982. However, due to insufficient consideration of helium's characteristics, the outcomes fell short of expectations, resulting in inadequate power output at the Energieversorgung Oberhausen plant (EVO) [1]. From 1997 to 2008, the Japan Atomic Energy Agency (JAEA) undertook the design and testing of a series of helium compressors (Figure 9). They analyzed the sensitivity of helium compressors to Reynolds numbers and established the relationship between variable efficiency and Reynolds numbers. Taking into account the properties of helium, Muto *et al*. [46] designed and experimentally validated helium turbines, high-pressure, and low-pressure centrifugal compressors. They obtained the aerodynamic performance of the helium turbine and compressor, established the relationship between variable efficiency and Reynolds numbers, and conducted experimental work on a four-stage helium compressor, revealing a close match between actual results and the design performance. In 2008, the Korea Advanced Institute of Science and Technology (KAIST), incorporating design methods

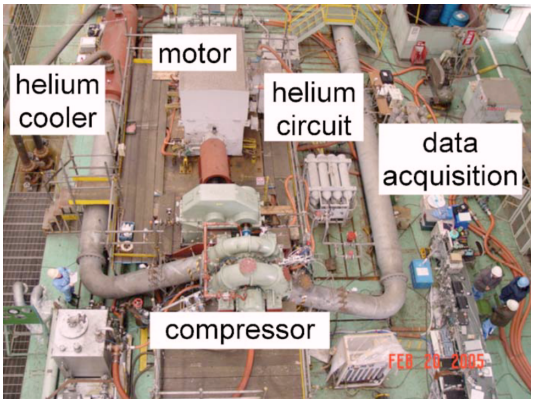


Figure 9 JAEA helium compressor experiment base [1]. Copyright 2008 by ASME.

utilized for air compressors, redesigned the helium compressor for the GTHTR300 [47]. The performance of the helium compressor, as determined by KAIST in terms of flow rate, closely aligned with the JAEA's design results. However, the efficiency of the compressor designed by KAIST far exceeded that of the compressor designed by JAEA at low speeds.

Yan and Lidsky [48] based on design methods such as gas turbines and considering the properties of helium, improved and designed a closed-cycle helium turbine suitable for modular high-temperature gas-cooled reactors. This turbine can achieve a grid efficiency of 45% at a core outlet temperature of 850°C, with the potential to exceed 50% efficiency at 950°C and above.

Chen *et al*. [49], utilizing dimensional analysis and velocity triangle analysis, proposed a similarity method composed of similarity criteria and a performance parameter transfer method. This method allows the performance characteristics of helium compressors to be obtained based on performance tests of air compressors. A comparison of the results was obtained using the similarity method and numerical simulations demonstrated high accuracy.

Using a 15 g mol^{-1} helium-xenon mixed gas as the working fluid, Malik *et al*. [50] designed a two-stage high-load axial flow compressor for a 300 MW high-temperature gas-cooled nuclear power plant. The number of stages is only 20% of that of the helium compressor, which significantly reduces the equipment size.

In the closed Brayton cycle system designed for space microreactor applications, the properties of helium-xenon gas may necessitate modifications to the number of turbine stages. Radial turbines and centrifugal compressors offer benefits in terms of compactness and weight. Low-power radial turbines, in particular, require high rotational speeds.

Gallo and El-Genk [51] conducted a series of studies on the Brayton cycle rotating unit (BRU) (Figure 10). They performed design selection and analysis for radial turbines and centrifugal compressors using helium-xenon gases with molecular weights of 15 and 40 g mol^{-1}, coupled with performance calculations for the S4 reactor space cycle system. El-Genk and Tournier [33,52,53] also studied helium turbines for small high-temperature gas-cooled reactors and investigated the effects of the working fluid, turbine inlet temperature, and shaft speed on the turbine size, number of stages, and performance.

Yuan *et al*. [54] conducted a design study on radial turbines applied to closed Brayton cycles in space. The working performance of four different working fluids (helium-xenon, helium, argon, and air) in radial

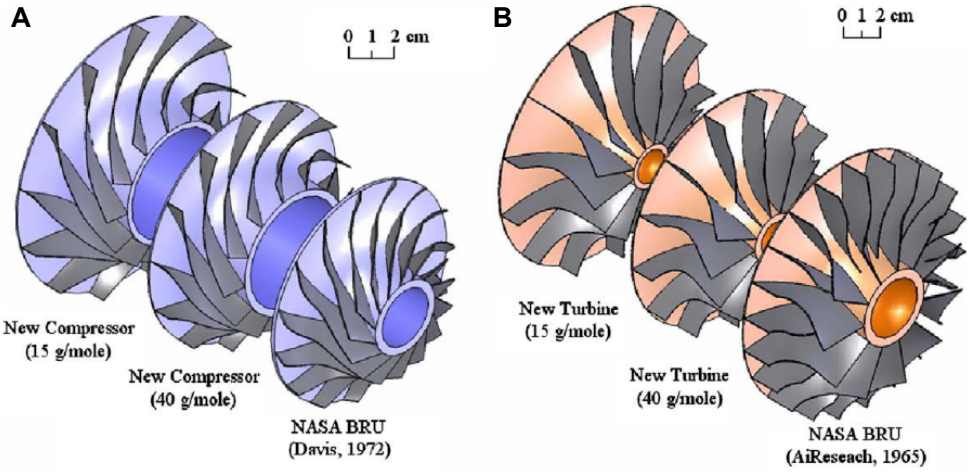


Figure 10 Helium turbine and compressor design [51]. Copyright 2009, Elsevier Ltd.

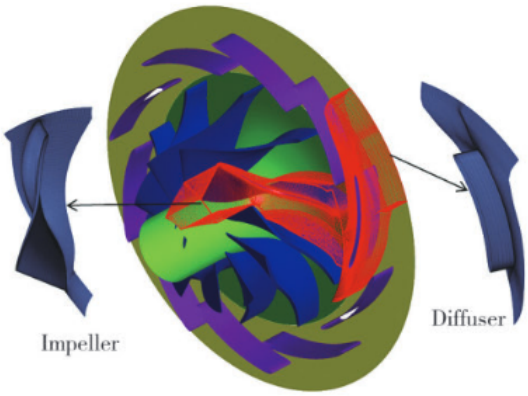


Figure 11 Helium-xenon centrifugal compressor design [58].

turbines was analyzed. The properties of helium-xenon with molecular weights of 15.9 and 40 g mol^{-1} were calculated using the Chapman-Enskog kinetic theory. The influence of parameters such as the Euler number, Reynolds number, and inflow velocity on the flat plate flow was considered. Yuan *et al*. [55] also analyzed the geometric parameters of the volute casing for helium-xenon radial turbines, and evaluated the casing performance in terms of losses and turbine inlet angle matching. The results showed that with a larger section angle, the casing sensitivity to changes in the angle decreased, and the overall losses decreased with increasing section angle. The throat loss of the casing was mainly due to the wake, and the exit flow angle decreased with increasing section angle. The exit angle was strongly sensitive to the exit width, and decreased as the width decreased.

Liu [56] designed a centrifugal compressor suitable for helium-xenon mixed working fluids and obtained selection rules for the design parameters of a helium-xenon centrifugal compressor. Tian [57] and Tian *et al*. [58] designed a centrifugal compressor for a helium-xenon mixture with a molar mass of 40 g mol^{-1} (Figure 11). The compressor had a single-stage total pressure ratio of 2.3, an isentropic efficiency of 88.7%, a flow rate of 1.6 kg s^{-1}, and a surge margin of 20.4%. Numerical simulations were performed to analyze the internal



Figure 12 High-temperature turbine components during high-power generation [63].

flow field of the compressor, resulting in characteristic curves for the helium-xenon centrifugal compressor.

Xu *et al*. [59,60] designed and numerically simulated a radial turbine for helium-xenon working fluid and investigated the effects of the outlet backpressure, tip clearance, and blade number on the turbine aerodynamic performance and flow field. They achieved a design with a total pressure ratio of 2.26, an efficiency of 84.4%, and a power of 618.3 kW. The analysis concluded that increasing outlet backpressure reduced flow rate and linearly decreased the output power of the turbine. The tip clearance should be as small as possible, and the number of the blades should be minimized, although too few blades lead to significant losses and lower efficiency.

Zhang [61] and Zhang *et al*. [62] optimized the design of a centrifugal compressor, specifically focusing on the impact of helium-xenon mixed gas properties and the presence of bypass blades on aerodynamic design and compressor performance. Alternative experimental tests were performed, substituting helium-xenon gas with argon-air gas. The alternative experimental approach was proved to be highly feasible, as the efficiency of both the argon and air compressors was found to be essentially equivalent to that of the helium-xenon compressor. This indicates that the alternative compressors can effectively reflect the efficiency characteristics required.

In March 2023, the independently developed space closed Brayton thermoelectric conversion system based on a helium-xenon mixed working fluid by the CASC Academy of Space Technology (801 Institute) underwent multiple successful system-level thermal tests (Figure 12). Through comprehensive parameter tuning, the system was subjected to variable operating conditions, achieving successful multi-level electric power outputs ranging from hundreds to thousands of kilowatts [63].

The researches of turbomachinery design and analysis are introduced in this part. Though several helium turbomachines are developed based on the traditional turbine and compressor, researches on more efficient helium-xenon turbomachines are mainly conducted by numerical simulation. In the future, manufacture on helium-xenon centrifugal compressor and radial turbine can be carried out.

Heat exchanger

Space Brayton cycle systems have high mass and volume requirements. Among the components occupying a significant portion of the system mass and volume, heat exchangers require careful selection and design.

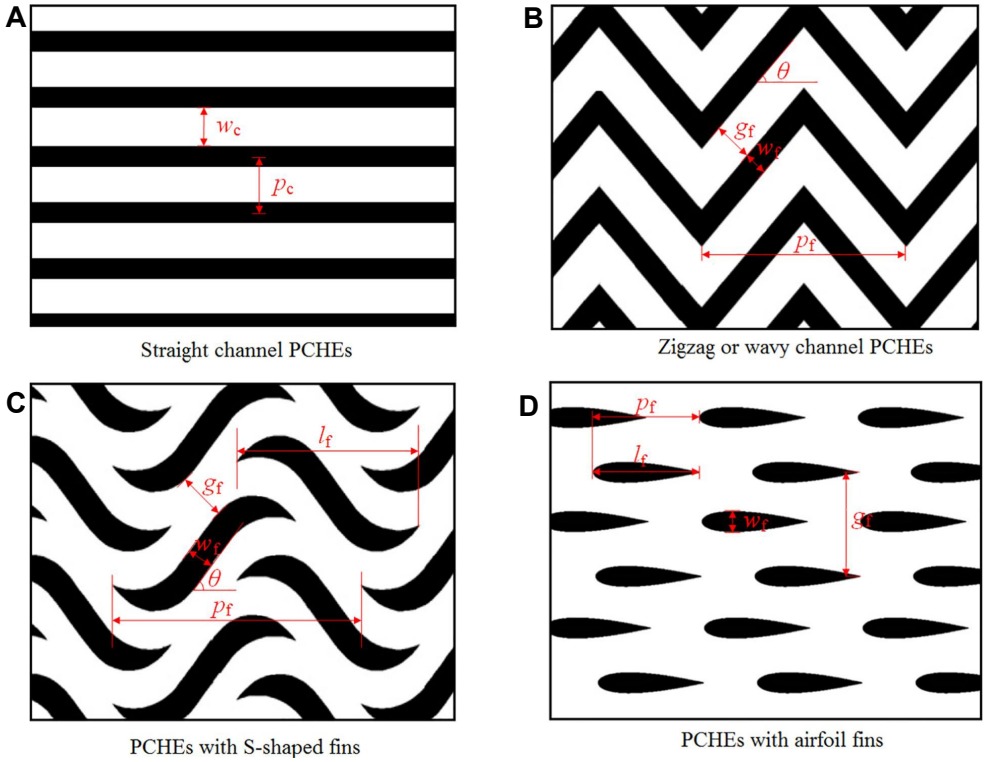


Figure 13 Four types of PCHE. (A) Straight channel, (B) zigzag/wavy channel, (C) S-shaped fins, (D) airfoil fins [64].

Primary Surface Heat Exchangers (PSHE) possess advantages such as a small volume, light weight, high recuperation efficiency, and strong reliability. These components are well-suited for the recuperating requirements of miniature reactors in helium-xenon Brayton cycles. Additionally, they can be produced through one-time stamping, facilitating manufacturing and reducing construction costs. Printed Circuit Heat Exchangers (PCHE) exhibit advantages such as high pressure and high-temperature resistance, compact structure, and high heat transfer density, making them suitable for deployment as heat exchangers in helium-xenon Brayton cycles (Figure 13) [64].

Song *et al*. [65] conducted numerical simulations and experimental studies on different types of primary surface heat exchangers. They established a computational model for a 120° staggered unit and performed numerical simulations of flow and heat transfer. They designed experiments and integrated a set of designs, material selection, and processing methods.

Ma *et al*. [66] comprehensively analyzed the heat transfer resistance characteristics of reheaters and established a closed Brayton cycle electric efficiency and specific working model for low-Prandtl number working fluids. They investigated the effects of the temperature ratio, pressure ratio, working fluid composition, and other parameters on the system performance. The existence of an optimal recuperation temperature was demonstrated, and the theoretical limits of the closed Brayton cycle efficiency and specific work were calculated.

Chai and Tassou [64] summarized the research progress on PCHEs in helium and S-CO_2 cycles. They introduced and summarized the selection of materials, manufacturing methods, channel types, heat transfer

flow characteristics, and design optimization methods for PCHEs. In general, wing-shaped fins exhibit the best heat transfer performance, followed by S-shaped fins and zigzag channel-type fins, whereas straight channel-type fins exhibit the poorest heat transfer performance. Kim *et al*. [67] and Kim and No [68] used computational fluid dynamics (CFD) simulations to study the thermal-hydraulic characteristics of wavy-PCHEs under helium-water conditions in both horizontal and vertical arrangements. They conducted experiments in the laminar flow region and found that when the PCHE is horizontally installed, the fluid flow becomes uneven, and local evaporation issues arise. Therefore, they recommended vertical installation for stable operation. Mylavarapu *et al*. [69] conducted thermal-hydraulic experiments on PCHEs with circular and semi-circular channels and calculated the friction and heat transfer characteristics of the heat transfer surface.

Geschke [70] developed a simulation tool for estimating the size and cost of heat exchangers. Using this tool, they compared energy conversion systems for microreactors. The volume of the helium-air heat exchanger was approximately 13.7 m^3, with a cost of $2.9 million. In contrast, the helium-water heat exchanger had a volume of 0.91 m^3 and a cost of $195,000, which is much greater than that of sodium-water or sodium-CO_2 heat exchangers. Therefore, optimizing the volume and mass of a heat exchanger can satisfy the requirements of space missions and reduce costs.

Yin *et al*. [71] modeled and analyzed the PCHE in the Small Innovative Helium-Xenon Cooled Mobile Nuclear Power Systems (SIMONS) and conducted optimization. They analyzed the geometric configuration of zigzag-type PCHEs, evaluated the influence of key geometric variables on the flow and heat transfer characteristics, and performed multi objective optimization to obtain the Pareto frontiers of the heat exchanger mass and heat transfer performance.

Yang and Huo [72] studied the impact factors such as temperature, recuperation, and manufacturing process on the performance of a helium-xenon cycle heat exchanger. They found that using precision machining could effectively reduce the heat exchanger mass by approximately 17%, and decrease the channel pressure drop by approximately 30%, while the heat transfer performance of the heat exchangers did not significantly decrease. They proposed using the area-specific power factor as an evaluation index for the heat exchanger structure and performance and conducted optimization based on this factor.

Shi *et al*. [73] conducted relevant research on the enhancement effect of helium-xenon heat transfer performance. They examined the impact of three different built-in vortex generators (a rectangular ring rib, a Kagome lattice, and a body-centered cubic (BCC) lattice) and their configurations on the flow and heat transfer performance of helium-xenon mixed gas. The results revealed that the three vortex generators exhibited distinct trends and performances in terms of heat transfer enhancement. Among them, the rectangular ring rib vortex generator demonstrated the most significant improvement in heat transfer, with a Nusselt number value 2.6 times higher than that of a smooth tube. The heat transfer enhancement phenomenon was observed specifically at the position of vortex generator. The BCC lattice vortex generator, which achieved a balance between heat transfer enhancement and pressure drop, exhibited the best overall performance, with an efficiency greater than the first one. Kagome and BCC lattice vortex generators possessed similar topological structures, as well as comparable flow and heat transfer characteristics. Both generators generated vortices at the upper end, resulting in a minor pressure drop while enhancing heat transfer. The addition of vortex generators effectively strengthened heat transfer but also increased the pressure drop. Calculations indicated that the tube equipped with 9 vortex generators provided the most

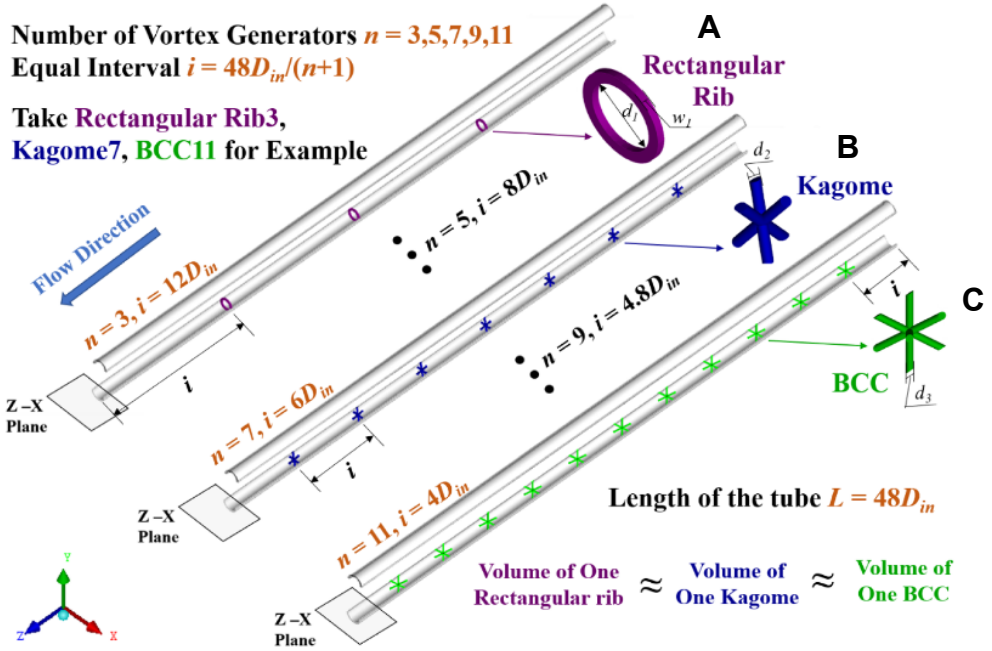


Figure 14 Physical models scheme. (A) Rectangular rib unit, (B) kagome unit, and (C) BCC unit [73]. Copyright 2023, Elsevier Ltd.

optimal comprehensive heat transfer performance (Figure 14).

In this part different kinds of heat exchangers and their heat transfer performance are analyzed and compared. In the future, performance concerning helium-xenon properties and manufacture of heat exchangers can be carried out.

Cooler

Owing to the unique characteristics of the space environment, the dissipation of waste heat in space reactors relies on radiative cooling. Therefore, the radiative cooler is an essential component in helium and helium-xenon cycles, and the design focus of the cold end, which primarily comprises the radiator, cooling medium, and radiative cooler, is one of the key research directions in helium-xenon cycle systems.

Since the 1990s, NASA has initiated research on high-performance space radiators, developing various novel heat pipe radiators, including single-channel heat pipe radiators, lightweight controllable heat pipe radiators, high-temperature heat pipe radiators, and heat pipe radiators for precision satellite thermal control. Generally, high-temperature heat pipes use metals such as liquid sodium, potassium, sodium-potassium, and lithium as working fluids, whereas low-temperature heat pipes typically use water as the working fluid.

El-Genk [74] conducted relevant design research on high-temperature water heat pipe radiators. They designed a set of radiators with six deployable panels for the three circuits of the S4 reactor. Heat was transported through a sodium-potassium loop, and water heat pipes with integral C-C fins were used to dissipate heat into space. The addition of C-C fins enhanced heat transfer but increased the mass of the radiator.

Romano and Ribeiro [75,76], operating in the cold-end temperature range of 450–500 K, utilized rec-

tangular groove titanium-water heat pipes as space radiators. They optimized these heat pipes to achieve a power-to-mass ratio. The mass ratio of the heat pipes ranged from 11 to 13 kg kW^{-1}, and the entire cycle system was expected to be within 40–50 kg kW^{-1}.

Liquid droplet radiators (LDRs) are another highly promising type of space radiator. In high-power closed Brayton cycle systems, where more heat needs to be dissipated, LDRs have a smaller volume and mass than heat pipe radiators. According to NASA's theoretical research, the unit mass heat dissipation power of LDRs can reach 250–450 W kg^{-1} [77]. Mattick and Hertzberg [78] conducted preliminary research on LDRs, indicating their potential to reduce the weight of space nuclear power cycle systems and their applicability in various space cooling applications.

Ma [79] conducted numerical simulations of the radiation and evaporation characteristics of LDRs. They established numerical models for the radiation-evaporation process of individual droplets and droplet clusters, developed evaporation transfer equations, constructed a radiation-evaporation coupled model, and discussed the effects of the initial droplet temperature and optical thickness on the temperature distribution, evaporation loss rate, and system lifetime.

Zeng [80] focused on the heat transfer performance of LDRs and studied the evaporation and radiation characteristics of a sparse-type LDR system. They developed a new simplified calculation method for a droplet layer radiation heat transfer model, which contributes to the design and development of LDRs.

Qin *et al*. [81] conducted a comparative study on the cycle performance of space gas-cooled reactors coupled with heat pipe radiators and LDRs. They found that the area and mass of the heat pipe radiator linearly increased with radiative heat dissipation. Under the conditions of 0.5 MW_e electrical power and a cycle inlet temperature of 1500 K, the mass of the LDR was only 10% of that of the heat pipe radiator, demonstrating significant advantages.

In summary, various countries have made considerable progress in research on the key components of helium and helium-xenon Brayton cycle systems. Relevant experiments were successfully conducted, laying a solid foundation for future comprehensive applications. However, it is still a great challenge for the helium-xenon heat transfer components due to the high cost and experimental difficulty. Moreover, the safety and long-time stable running of the components is also the target of future research.

DYNAMIC ANALYSIS AND CONTROL STRATEGY

Dynamic analysis

Researching the dynamic characteristics of helium-xenon Brayton cycle systems and establishing efficient and flexible control strategies are of significant importance for ensuring the safe and long-term operation of these systems. Researchers worldwide have established dynamic simulation models for specific systems, studied control strategies, and conducted dynamic investigations numerically. Specific classifications and introductions are provided in the section of Specific conditions.

Investigations on dynamic characteristics of helium-xenon Brayton cycle systems are conducted mainly by establishing the dynamic models of the system, reactor and components via computer program. Using the dynamic models, researchers conducted detailed research on dynamic processes such as transient start-up, shut-down, load variations, and accident conditions.

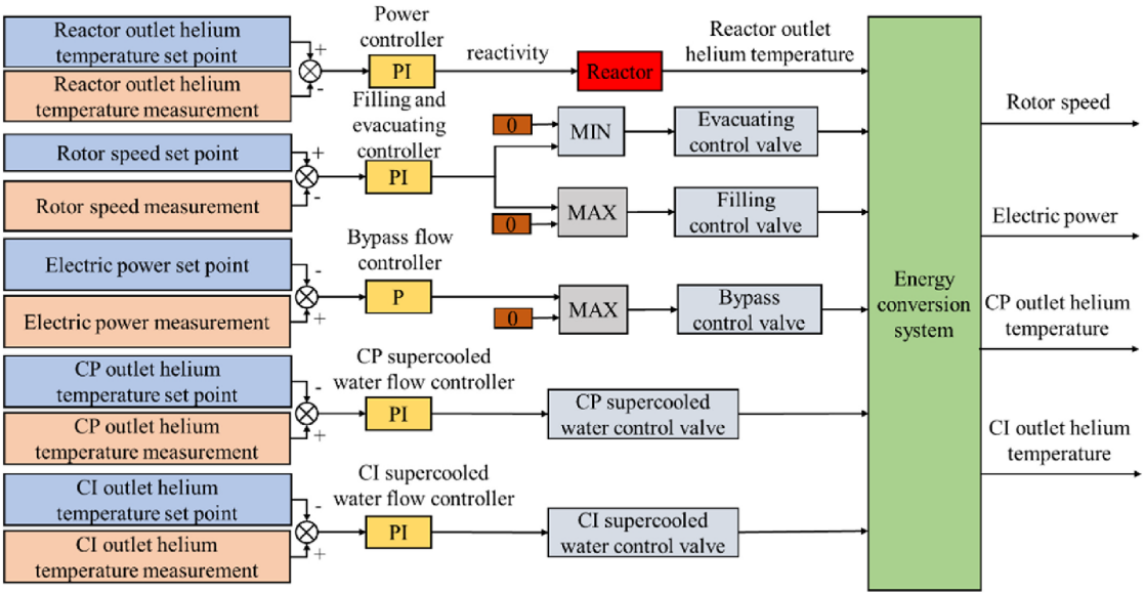


Figure 15 Mi-HTR control system scheme [86]. Copyright 2023, Elsevier Ltd.

Huang and Feng [82,83] from the Nuclear Energy Research Institute of Tsinghua University established a two-dimensional dynamic model of the HTR-10 high-temperature gas-cooled reactor, and conducted a dynamic simulation of the reactor core to quickly react for dynamic simulation. Ma *et al*. [84] developed an aggregated parameter mathematical model for various components of a helium gas turbine in a high-temperature gas-cooled reactor. Using MATLAB/Simulink simulation software, they constructed a simulation model to study the transient operational characteristics during the startup process of a helium gas turbine, obtaining the dynamic response of key parameters during the startup process with respect to the reactor core power.

Using the Simulink platform, Chi [85] simulated the startup and variable operating conditions of a space lithium-cooled reactor coupled with a helium-xenon Brayton cycle system. This study explored the dynamic response analysis of the system during startup, at variable speeds and variable loads, and during the introduction of reactive perturbations. The results demonstrated that the system possesses certain negative feedback regulation capabilities, enabling the maintenance of normal system operation.

Qiu *et al*. [86] modeled a dynamic control system for a micro size high-temperature gas-cooled reactor with a helium-xenon Brayton cycle. They analyzed disturbances such as the reactor power, external load, and flow rate to obtain a dynamic characteristic model with characteristics such as large time constants, nonlinearity, strong coupling, and time-varying parameters. The designed control system demonstrated the ability to respond to step changes in the external load by up to 33% (Figure 15).

In experimental studies, owing to the high cost of helium and helium-xenon mixed gases, alternative experiments often use air or nitrogen as substitutes. For example, the MAGNET system at Idaho National Laboratory employs air or nitrogen as the working fluid [87]. Given the inherent risks associated with the dynamic characteristics of microreactors, the MAGNET system utilizes electric heating to simulate the reactor heat source. This approach allows for testing under steady-state operation, variable conditions, single-point faults, and other scenarios while avoiding the impact of nuclear radiation on experimental personnel.

Table 5 Control strategies of the helium Brayton cycle

Control method	Characteristic
Bypass valve control	(1) Different positions can be selected; load change process is smooth. (2) Need high temperature helium valves.
Inventory tank control	(1) High efficiency, but slow changing speed. (2) Limited by tank inventory.
Variable guide vane control	(1) Wide range of variable load adjustment. (2) Need expensive high temperature variable guide vane.

Control method

Scholars from all over the world have conducted relevant research on the dynamic characteristics of helium Brayton cycles. However, in comparison to supercritical carbon dioxide cycles, the control strategies of closed helium Brayton cycles have been the subjects of relatively less research. A summary of previous studies revealed that bypass valve control and reservoir control were the primary control methods (Table 5).

In research on the control strategies for helium Brayton cycles, Xie and Wang [88] utilized MATLAB to develop a dynamic calculation program for a helium turbine closed cycle system. This program was used to analyze the variations in parameters such as the helium flow rate, temperature, speed, power, and compressor surge margin in the event of a grid load loss accident and a transient condition with a 5% reduction in the heat transfer of the main heat exchanger. To maintain system safety and stability during power changes, bypass circuits, safety valves, and control valves were installed between the turbine outlet and the high-pressure compressor outlet. The results indicated that, in the case of a grid load loss accident, the compressor and turbine could return to a safe state by adjusting the bypass circuit valve. Under transient conditions with a 5% reduction in the heat transfer of the main heat exchanger, the control valve opening scheme was most effective at maintaining the turbine inlet temperature, whereas the low-opening scheme resulted in minimal shaft power variation.

Li *et al*. [89] from Tsinghua University investigated the control strategy of the helium recuperating Brayton cycle with a focus on bypass valve control. The dynamic characteristics of the system were clarified by studying the impact of single and double bypass valve opening at different locations, and optimization was conducted for the selection of single or double valves. Combined with proportional integral derivative (PID) control, this strategy effectively reduced coupling and enhanced system robustness.

Ma *et al*. [90] studied the rapid power regulation characteristics of a space reactor system under bypass valve control and compared the sensitivity effects of different bypass valve openings. The results indicated that changing the bypass valve opening altered the pressure and flow at various nodes in the system, thereby modifying the mechanical conditions of the turbine. The speed of the turbine compressor increased with decreasing cycle load. Increasing the bypass valve opening allowed timely control of the shaft speed. After increasing the bypass valve opening, the pressure on the low-pressure side of the pipeline components increased, the temperature of the cooling cycle increased, and the radiator required greater heat dissipation capacity.

Hao *et al*. [91] conducted an analysis of the dynamic response characteristics of HTR-10GT, improving the models of key components and introducing the influence of volume inertia. They analyzed the dynamic characteristics of the system during bypass valve regulation and provided a mechanism for reducing the

Table 6 Control options [93]

No.	Option
1	Power turbine bypass and power turbine inline valves
2	Variable guide vanes on all the turbines
3	Main compressor bypass with variable guide vanes on all the turbines
4	Variable guide vanes on the HP and LP turbines together with a power turbine bypass valve
5	Compressor bypass valve only
6	Recuperator bypass valve with a resistor bank
7	Compressor bypass together with variable guide vanes on the HP and LP compressors
8	Compressor bypass and local bypass valves over the HP and LP compressors

output power. The results showed that the volume inertia of the system had a significant impact on the response speed of the bypass valve regulation, and that the valve opening determined the output power of the system in the final state. Temperature shock in the reheater might occur during the regulation process but could be mitigated by the coordinated operation of the two bypass valves. The change in the outlet temperature of the reactor was minimal; therefore, the accuracy of the reactor model had little effect on the results.

Li [14] conducted a study on the dynamic characteristics and control strategies of a space reactor helium-xenon Brayton cycle system under variable load conditions. Three control strategies are used: system filling control, bypass regulation, and variable speed control. The results showed that under variable load conditions, the filling level adjustment and bypass ratio adjustment could maintain a constant system speed. Filling level control could maintain a high cycle efficiency, while bypass regulation, although allowing for rapid load changes, had a narrow load adjustment range. The principle of variable speed control is to deviate the system from the design point by changing the compressor speed, thereby reducing the output power of the system. In the event of an accident, because the system load shedding speed increases rapidly, bypass regulation can be used to quickly reduce the output power and reduce the system speed to a safe range.

Gad-Briggs *et al*. [92] studied the load control strategy of simple recuperative (SCR) and intermediate cooling (ICR) helium Brayton cycles, mainly using inventory pressure control for load adjustment. Inventory pressure control was employed to adjust the mass flow rate, thereby changing the pressure level in the helium circuit without altering the speed or pressure ratio of the compressor. In comparison, the ICR cycle required 102% longer than the SCR cycle to reduce the load by 50%. In the ICR cycle, any attempt to adjust the flow rate of the control valve must consider the aerodynamic stability of the compressor to avoid surge.

Botha and Rousseau [93] investigated the load rejection control strategy of a three-shaft intercooled closed Brayton cycle with a high-temperature gas-cooled reactor as the heat source and helium as the coolant. They compared the control effects of eight different combination control strategies, and the advantages and disadvantages of each control method are summarized in Table 6. The optimal control methods were presented in 8 schemes. Scheme 1, utilizing a power turbine bypass and a power turbine control valve, provided a stable method for controlling the system during load rejection and could be used at full load. The valves were required to operate in a high-temperature helium environment; another drawback was the need for a complex mixing chamber. Scheme 8, which used a compressor bypass valve, had the main advantage that it could be based on existing valve technology because the operating temperature of the valve was 150°C, but it could not achieve full load control.

Specific conditions

Load variation

Wang *et al*. [94] developed a physical property calculation model for helium-xenon mixed gas based on semiempirical formulas. They established a simulation program for a closed Brayton cycle system and conducted a steady-state response analysis, transient calculations for positive and negative reactivity insertions, and reactor step-load reductions. The error compared with the design values did not exceed 5%, indicating that the simulation accurately reflected the operational results.

Ming *et al*. [95] focused on establishing a dynamic simulation program for a direct Brayton cycle reactor system. The results showed high accuracy, with a relative error compared to the design values, which did not exceed 6.38%. This program can effectively reflect the changing trends of the key parameters in the system under various operating conditions in real-time.

Hou [96] utilized the RELAP5 program to construct a dynamic model for a lithium-cooled fast reactor coupled with a helium-xenon Brayton cycle system. The dynamic simulation included scenarios such as power increase and decrease, criticality safety and flow loss accidents, reactivity insertion accidents, and heat trap loss accidents in space nuclear reactor systems. This model can be employed for safety analysis and the study of transient operational characteristics. The simulation results demonstrated that the system could respond rapidly, and all the parameters remained within the safety limits during various accident conditions.

Start-up and shut-down

El-Genk *et al*. [97] established a dynamic simulation model for the S4-CBC space power system using the Simulink platform. They simulated the transient response during startup and demonstrated the startup process of a Brayton cycle system with thermal power of 471 kW_{th}, turbine speed of 45 kr min^{-1}, turbine inlet temperature of 1149 K, and compressor inlet temperature of 400 K. The simulation results showed a startup time of 4 h, electrical power of 130.8 kW_e, and thermal efficiency of 27.8%, confirming the transient capability of the system. This model can be applied to simulate the transient operation process of other closed Brayton cycle energy conversion space reactor power systems.

Wang *et al*. [98] used the Brayton Cycle Reactor System Analysis Program (BRESA) analysis program to dynamically simulate the heat removal of a helium-xenon Brayton cycle system after reactor shutdown. The results indicated that assuming a residual power of 10% after reactor shutdown and maintaining the original power of the cooler, the system's natural circulation capability could remove residual heat from the reactor core within 1500 s. When the cooler power was reduced to 50%, the system discharged the residual heat within 4000 s. In the event of a reactor scram with cooler failure, it was estimated that the core temperature would exceed 1200 K after 553 s, allowing sufficient time for manual intervention.

Accidents

Zhang *et al*. [99] conducted research on a megawatt-level gas-cooled space reactor coupled with a Brayton cycle power system. They developed an integrated dynamic model encompassing the reactor, turbine machinery, radiators, heat exchangers, and pipes. This research included the establishment of a transient

response analysis program for accident scenarios such as single-loop failure and reactivity insertion. Wang *et al*. [100] utilized this program to analyze the dynamic processes during startup, reactivity insertion accidents, and residual heat removal after shutdown. Qin *et al*. [101] employed CFD to create an open-grid MEGAwatt-class space nuclear reactor system (OMEGA) and analyzed transient processes related to reactivity insertion and partial coolant loss.

Zhou *et al*. [102] analyzed initiating events for SIMONS by employing the Main Logic Diagram (MLD) analysis method to identify 31 initiating events categorized into six groups: core heat removal increase, core heat removal decrease, abnormal reactions and power distribution, pipe cracks and equipment leaks, transient conditions without an emergency scram, and internal and external disasters. The results obtained can serve as a theoretical basis for further safety analyses of SIMONS.

Using probabilistic safety assessment, Wu *et al*. [103] conducted a frequency analysis of severe accidents in a small helium-xenon-cooled mobile reactor power source based on operational experience and component failure data from high-temperature gas-cooled reactors and pressurized water reactors. This study provides valuable references for reactor design improvements, fault diagnosis, and operational guidance.

Generally, dynamic research on the effects of helium and helium-xenon Brayton cycle has made some progress, and the control methods of the existing air or CO_2 Brayton cycle can be used as references. But the research is still limited in the stage of theoretical simulation because of high cost and operation. More experiments will be required in the future. Besides, some control methods are not appropriate for space helium-xenon systems, which need more adjustments.

CONCLUSION

Brayton cycles, utilizing helium-xenon mixed gas as the working fluid, have significant potential as an energy conversion system for small nuclear reactors in remote land, deep-sea, and space applications. This review conducts an in-depth investigation of the research progress on helium-xenon Brayton cycle systems tailored for small nuclear reactors and provides a comprehensive review of related advancements.

The heat transfer and flow characteristics of the working fluid are influenced by the different ratios of helium and xenon in the mixture. Currently, mainstream designs primarily use the helium-xenon mixed gas ratios of 40 and 15 g mol^{-1}. The 40 g mol^{-1} mixture exhibits excellent heat transfer performance and superior compression characteristics, allowing for a reduction in the compressor stage and overall system mass, allowing it to be widely adopted in space microreactor Brayton cycles. The 15 g mol^{-1} mixture has a higher heat transfer coefficient by 6%–7% than helium alone, with turbine mechanical stages of only 24%–30% of those for helium. However, it experiences significant pressure losses during the flow process, making it suitable for terrestrial Brayton cycles. Despite the prevalent theoretical analyses and initiatory experimental investigations of helium-xenon gas properties, future research should integrate experimental analyses and turbomachine performance to enhance the understanding of the actual gas properties in Brayton systems. Moreover, there is a need for targeted discussion on component structure design and flow heat transfer issues at different working fluid ratios. Future studies should determine specific helium-xenon gas mix ratios for different power levels, application scenarios, and system configurations, and conduct further research on the characteristics of the working fluid under these ratios.

The optimization and analysis of the system layout and parameters are crucial aspects of helium-xenon Brayton cycle design research. Large high-temperature gas-cooled nuclear power plants typically adopt an intercooling and recuperating configuration to achieve a higher cycle efficiency. In contrast, mobile microreactors matched with helium-xenon cycles often use a simple recuperating configuration to reduce the overall size and improve the compactness of the system. High efficiency and compactness are typically targeted during the design of helium-xenon Brayton cycles. However, more detailed layout designs are still in demand including component position and multi-loop cycle system compact design. Optimization revolves around parameters such as the cycle pressure ratio, turbine inlet temperature, and recuperator effectiveness, analyzing their impact on system performance metrics such as energy conversion efficiency, thermal efficiency, power density, and power-to-mass ratio. However, determining the relative importance of each performance metric in multi-objective optimization and achieving optimal decision-making requires scenario-specific analysis. Currently, there is limited research on the comprehensive consideration of power variations and cold/hot source conditions, warranting future research on the integrated optimization of coupled reactors and cold-end systems.

The key components of the helium-xenon Brayton cycle include turbines, compressors, and heat exchange equipment such as the main heat exchanger, reheater, and cooler. To achieve a compact turbine design, considering the pressure ratio and flow, a single-stage or multi-stage radial inflow turbine and compressor can be employed. High-temperature, high-pressure-resistant, compact, and high-performance PCHEs or PSHEs are suitable options for recuperators. The design of the main heat exchanger depends on the reactor type and layout, with helium-xenon Brayton cycles typically matched with high-temperature gas-cooled reactors or heat pipe-cooled reactors. Space helium-xenon Brayton cycle systems often use heat pipe radiators or droplet radiative coolers for thermal dissipation. However, current calculations for turbines and heat exchangers involve varying degrees of simplification, relying on existing literature data and empirical formulas, without fully considering real-world applications and working fluid properties. Future research can refine the modeling by considering the working performance of turbines and heat exchangers under different conditions and working fluid ratios, and analyzing parameters such as pressure loss, variable efficiency, and heat exchange efficiency, leading to more mature helium-xenon cycle system component design solutions. Subsequent experimental studies can then be conducted based on these solutions.

Simulation calculations for the dynamic processes of helium-xenon cycles under load variations and accident conditions and the design of control strategies based on the results are crucial. Previous research has simulated processes associated with system start-up and shut-down, load variation, and accident conditions such as reactivity insertion, criticality, loss of flow, breach, and emergency shutdown. However, due to limitations in terms of environment, equipment, and cost, experimental research on the dynamic characteristics of helium-xenon Brayton cycles remains limited. Current experimental studies often employ alternative methods, such as substituting electric heating for reactor heating or using substitute fluids with similar properties (e.g., air, nitrogen, argon). Future research can further investigate the similarities of substitute fluids and develop empirical formulas that are more accurate and applicable to a wider range. The design of control strategies for helium-xenon Brayton cycles integrated with reactors and cold-end systems is still in its early stages, and future research can focus on dynamic response studies of coupled systems, proposing comprehensive dynamic control strategies that prioritize safety, flexibility, and efficiency.

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W.X.C designed the review, wrote, reviewed and edited the manuscript; M.H.S wrote, edited the manuscript, collected and summarized the literature; Y.R.Q reviewed, edited the manuscript and searched the literature; H.X.Y supervised the review; J. J.Y designed and supervised the review.

Conflict of interest

The authors declare no conflict of interest.

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